NRES 746, Week 11, Lecture 1: Linear or Not?

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

Instructions: Work alone on Part 1 (about 5 minutes), then compare with a neighbor for Part 2. Keep laptops closed.

Part 1: Sort the models (individual)

Here are six models for a response Y and a continuous predictor x. In (f), a_{site} means a separate intercept for each of three sites.

(a) $Y \sim \text{Normal}(a + bx + cx^2,\ \sigma^2)$

(b) $Y \sim \text{Normal}(ax^b,\ \sigma^2)$

(c) $Y \sim \text{Normal}(a + b \log x,\ \sigma^2)$

(d) $Y \sim \text{Poisson}(e^{a+bx})$

(e) $Y \sim \text{Normal}\left(\dfrac{ax}{b+x},\ \sigma^2\right)$

(f) $Y \sim \text{Normal}(a_{\text{site}} + bx,\ \sigma^2)$

1a. For each model, is the mean a linear function of the parameters (yes/no)? Could `lm()` fit it as written? Could `glm()`?

Model	Linear in the parameters?	lm()?	glm()?
(a)	*yes*	*yes*	*yes*
(b)	*no*	*no*	*no*
(c)	*yes*	*yes*	*yes*
(d)	*no (linear on the log scale)*	*no*	*yes*
(e)	*no*	*no*	*no*
(f)	*yes*	*yes*	*yes*

> (a), (c), and (f) are linear models; `glm()` fits them with `family = gaussian`. (d) is a Poisson GLM with a log link. (b) and (e) need `nls()` or a custom likelihood with `optim()`. (Strictly, (b) can be fit as a Normal GLM with a log link, since $\log \mu = \log a + b \log x$, and (e) as a Normal GLM with an inverse link, since $1/\mu = 1/a + (b/a)(1/x)$; "no" is fine for both.)

Part 2: Compare (with a neighbor)

2a. Taking logs of the mean in (b) gives $\log(ax^b) = \log a + b \log x$, which is linear. Does that mean `lm(log(y) ~ log(x))` fits model (b)? What would it assume about the noise?

> No. `lm(log(y) ~ log(x))` fits $\log Y \sim \text{Normal}(\log a + b \log x, \sigma^2)$, so Y is lognormal: the noise is multiplicative, its spread grows with the mean, and ax^b becomes the median of Y rather than its mean. Model (b) has constant, additive Normal noise. They are different models and can give different estimates of a and b.